

Chapter 1 Fundamentals

1.1 Definitions

1.1.1 Categories

1.1.2 Functors

Full, Faithful

Definition 1.1.2.1 A functor $F : X \rightarrow Y$

$$F_{X,Y} : Hom_C(X, Y) \rightarrow Hom_D(FX, FY)$$

is called

- **full** if surjective;
- **faithful** if injective;
- **faithfully full** if bijective;

Continuous Functors

Definition 1.1.2.2 A functor $F : C \rightarrow D$ is **continuous** if it preserves all **small limits** that exists in the domain C .

i.e. given any **small category diagram** $A : E \rightarrow C$ in C for which

- $\lim(A)$ exists;
- the universal cone $\lim(A) \rightarrow A$

Application of F to that cone results in a universal cone $F(\lim(A)) \rightarrow F \circ A$. Thus making $F(\lim(A))$ the limit of $F \circ A$.

The limit $\lim(F \circ A)$ exists, and the natural map induced by the universal property of the limit

$$F(\lim(A)) \rightarrow \lim(F \circ A)$$

is an isomorphism.

Continuous Functors in Enriched Category Theory

Definition 1.1.2.3 ...

1.1.3 Natural Transformation

Definition 1.1.3.1 Consider the natural transformation of two **natural isomorphism** if there exists functor $G : D \rightarrow C$ such that

$$GF = id_C \quad FG = id_D$$

◇ **Lemma 1.1.3.2** Consider a natural transformation $\eta : F \rightarrow G$, where $F, G : C \rightarrow D$.
 η is a natural isomorphism $\Leftrightarrow \eta(x) : F(x) \rightarrow G(x)$ are isomorphisms, $x \in X$.

◦ **Proof ...**

□

1.1.4 Adjunction

Definition 1.1.4.1 Let $\mathcal{C}, \mathcal{D}$ be categories. pair of functors

$$F : \mathcal{C} \rightarrow \mathcal{D} \quad G : \mathcal{D} \rightarrow \mathcal{C}$$

are called **adjunction**, if there exists a bijection

$$\Phi_{x,y} : \text{Hom}_{\mathcal{C}}(Fx, y) \rightarrow \text{Hom}_{\mathcal{D}}(x, Gy)$$

Unit;Counit

Definition 1.1.4.2 Let $\mathcal{C}, \mathcal{D}$ be categories, consider a pair of functors

$$F : \mathcal{C} \rightarrow \mathcal{D} \quad G : \mathcal{D} \rightarrow \mathcal{C}$$

then, the natural transformation

$$\epsilon : id_{\mathcal{C}} \rightarrow GF \quad \eta : FG \rightarrow id_{\mathcal{D}}$$

are called **unit** and **counit** of F, G , if together they satisfy

$$(\eta * F) \circ (F * \epsilon) = id_F$$

$$(\eta * G) \circ (G * \epsilon) = id_G$$

$$\begin{array}{ccc} & FGF & \\ F * \epsilon \nearrow & & \searrow \eta * F \\ F & \xrightarrow{id_F} & F \end{array} \quad \begin{array}{ccc} & GFG & \\ \epsilon * G \nearrow & & \searrow \eta * G \\ G & \xrightarrow{id_G} & G \end{array}$$

♠ Proposition 1.1.4.3 ...

◦ **Proof** ...

□

Adjoint Functor Theorem

► **Theorem** Sufficient Conditions for a Limit-preserving functor $R : \mathcal{C} \rightarrow \mathcal{D}$ to be a **right adjoint** include:

- (GAFT, General Adjoint Functor Theorem) $\mathcal{C}$ is **complete** and **locally small**, and R satisfies set condition;
- (SAFT, Special Adjoint Functor Theorem) $\mathcal{C}$ is **complete**, **locally small**, **well-powered**, and as a **small co-generating set**, and $\mathcal{D}$ is **locally small**. This

◦ **Proof** ...

□

► **Theorem** Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor between **locally presentable categories**, then

- F has a right adjoint $\Leftrightarrow$ It preserves all small colimits;

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1.1.5 Equivalence of Categories

'Principle of Equivalence'

<https://ncatlab.org/nlab/show/principle+of+equivalence>

Equivalence of Categories

Definition 1.1.5.1

- One object $A \in C$ is equivalent to another object $B \in C$, every context with A can be replaced by B .
- In higher category theory, an **equivalence** often means a morphism which is invertible in a maximally weak sense.

Two objects are **equivalent** if there is an equivalence between them.

Definition 1.1.5.2 Let $F : C \leftrightarrow D$ be an adjunction. The adjunction is called an **adjoint equivalence**, if the unit and counit are natural isomorphisms.

In this case, F and G are called **equivalences of categories**.

Colimits

1.1.6 Product Category

Definition 1.1.6.1 For C and D two categories. The **product category** $C \times D$ is the category:

- Objects are the **pairs**

$$(c, d) \quad c \in C, d \in D$$

written as $Obj(C \times D) = Obj(C) \times Obj(D)$;

- Morphisms

$$(f : c \rightarrow c', g : d \rightarrow d') \quad Mor(C \times D) = Mor(C) \times Mor(D)$$

- Composition of Morphisms, defined componentwise by composition in C and D .

◇ **Lemma 1.1.6.2** The following types are equivalent:

- Functors $A \times B \rightarrow C$;
- Functors $A \rightarrow C^B$.

◦ **Proof ...**

□

1.1.7 Generating Objects

Definition 1.1.7.1 Let C be a category. An object G is called a **generator** if it satisfies the following property: For $f, g : X \rightarrow Y$; $f \neq g$, then there exists a morphism from generator G ,

$$h : G \rightarrow X$$

such that $f \circ h \neq g \circ h$.

This implies, that the performance of h decides the performance of X .

1.2 Limits, Colimits (in Small Category)

1.2.1 Limits

Set-Theoretic Limits

Limit of Functors in an Arbitrary Category

Definition 1.2.1.1 (Limit of a functor, with values in an arbitrary category) Now, take the notion of limit for functors

$$F^{op} : D^{op} \rightarrow C$$

for C an arbitrary category, if we construct a certain presheaf C , which we will call $\hat{\lim} F$.
More precisely, using the **Yoneda embedding**

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Then:

$$(\hat{\lim})(c) := Hom_{Set^{D^{op}}}(pt, Hom_C(c, F(-)))$$

Universal Cone

Definition 1.2.1.2 (Universal Cone) Let $F : D^{op} \rightarrow C$ be a functor. Then, every object $c \in C$, an element

$$* \rightarrow Hom(pt, Hom(c, F(-)))$$

is identified with a collection of morphisms $(c \rightarrow F(d))$.
For all $d \in D$, such that all triangles

$$\begin{array}{ccc} & c & \\ & \downarrow & \\ F(d_i) & & F(d_j) \end{array}$$

Such a collection of morphisms is called a **cone** over F .

1.2.2 Colimits

1.2.3 Completeness, Co-Completeness

Definition 1.2.3.1 ...

Chapter 2 Presheaves, Yoneda Lemma

2.1 Presheaves

Definition 2.1.0.1 Let C be a category:

- A **presheaf** is a functor

$$\mathcal{F} : C^{op} \rightarrow Set$$

2.1.1 Yoneda Embedding

Definition 2.1.1.1 Define:

$$Y : C \rightarrow Psh(C)$$

With Hom -functor $Hom : C^{op} \times C \rightarrow Set$, there exists adjunction

◇ **Lemma 2.1.1.2** ...

◦ **Proof** ...

□

♠ **Proposition 2.1.1.3** Yoneda functor is:

- fully faithful;
- Preservation and Reflection the Isomorphisms;
- Dense(Density);
- Interaction With Density Comonads;
- Interaction With Codensity Monads;

◦ **Proof** ...

□

2.1.2 Yoneda's Lemma

► **Theorem** (Yoneda) Let $\mathcal{F} : C^{op} \rightarrow Set$ be a presheaf. We have following bijection

$$Nat(h_A, \mathcal{F}) \cong \mathcal{F}(A)$$

Natural in $A \in Obj(C)$. Especially, determining a natural isomorphism of functors

$$Nat(h_-, \mathcal{F}) \cong \mathcal{F}$$

◦ **Proof** ...

□

2.1.3 Properties of Categories of Presheaves

♠ Proposition 2.1.3.1 ...

◦ Proof ...

□

2.2 Representable Functors

2.2.1 Generalized Element

- <https://ncatlab.org/nlab/show/generalized+element>
- https://www.voutsadakis.com/TEACH/LECTURES/CATEGORY/Chapter4.pdf?utm_source=chatgpt.com P31

In category Set , the element can be seen as the maps

$$x : 1 \rightarrow A; A \in Ob(Set)$$

And in general

Definition 2.2.1.1 Let: C be an category and $A \in C$. Then, a morphism

$$x : S \rightarrow A$$

is a generalized element of A of shape S .

Especially, the generalized element is associated to the Yoneda embedding

$$Y : C \hookrightarrow [C^{op}, Set] \quad X \mapsto GenEl(X)$$

Under this embedding, $GenEl(X) : C^{op} \rightarrow Set, Ob(C) \ni U \mapsto (U \rightarrow X)$.

2.2.2 Locally Representable Category

Definition 2.2.2.1 ...

2.3 Copresheaf

2.4 Restsricted Yoneda Embedding, Yoneda Extension

Chapter 3 Monoid Category

3.1 Fundamentals

3.1.1 Moduli Category

Definition 3.1.1.1 A **monoidal category** is a category $\mathcal{C}$ equipped with

- A functor

$$\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$$

out of the product category of $\mathcal{C}$ with itself, called the **tensor product**.

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♣ Example 3.1.1.2

- Category $\mathbf{Vect}_k$;